# 12.17 Banach's Matchboxes

WORKSHEET

**The question**

A taxi driver keeps a packet of toffees and a packet of mints in her taxi. From time to time she takes either a toffee (with probability $p$) or mint (with probability $q = 1 - p$). At the beginning of the week she has $n$ toffees and $m$ mints in the packets. On the $N$th occasion that she reaches for a sweet, she discovers (for the first time) that she has run out of that kind of sweet. What is the probability that she was reaching for a toffee?

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

**Ask yourself** For the driver to discover on the $N$th reach that the toffees have run out, what must the $N$th reach be? What must the previous $N-1$ reaches contain?

### STAGE 1 Discovering the toffees are gone

**Ask yourself** In what order can the first $N-1$ reaches occur? What condition on $N-1-n$ ensures she never found the mint packet empty?

### STAGE 3 The conditional probability

**Ask yourself** We are told that a discovery happens on the $N$th reach. What is the probability that it was a toffee?

**Pause and try** Check the formula when $n=m$ and $p=q=\frac{1}{2}$. By symmetry the answer should be $\frac{1}{2}$ for every possible $N$. Is it?

## Try this next

1. **Small case.** With $n=m=1$ and $p=\frac{1}{2}$, list the possible sequences and find the distribution of $N$.
2. **Banach's formula.** Verify that $\sum_{k=0}^{n}\binom{2n-k}{n}2^{-(2n-k)}=1$ for $n=1$ and $n=2$.
3. **Equal tops.** Show that $\mathrm{P}(K=0)=\mathrm{P}(K=1)$ for every $n$ in Banach's problem.
4. **Which runs out first?** For the taxi driver, find the probability that the toffees are the first packet to be discovered empty (at any time), when $n=m=2$ and $p=\frac{2}{3}$.

## Further reading


- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter VI, which introduced Banach's problem to the world.
- **R. Daniel Mauldin (ed.), *The Scottish Book*** (Birkhäuser). The problems from the Scottish Café, with commentary.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University

Press), on the negative binomial distribution.

The full guided solution, and what the question leads to at university, is in Section 12.17:
https://keithmcnulty.github.io/step_book_mcnulty/ch12/matchboxes.html